Difference among Automation, AI Workflows and AI Agents

Automation

Definition:

A system that executes predefined, rule-based tasks automatically.

Core Characteristics:

- **Deterministic Logic:** Typically uses Boolean (true/false) or other clear, deterministic rules (e.g. if-then statements).
- **Fixed Decision Paths:** Operates within clearly defined parameters without ambiguity.

Additional Key Features:

- **Repeatability**: Processes are highly repeatable and predictable.
- **Minimal Variance**: Outcomes are consistent given the same inputs.
- **Real-Time Execution**: Often designed for timely, predictable responses.
- **Low Ongoing Maintenance**: Once configured and tested, the system generally requires little change unless the rules or environment change.

Note: While many automation systems rely on Boolean logic, some may incorporate more complex condition checks or even simple statistical rules. However, the key is that the logic is predetermined and not adaptive.

AI Workflow

Definition:

A process or system that integrates one or more AI components (for example, calling a Large Language Model [LLM] via an API) as part of a multi-step sequence to process inputs, make decisions, or generate outputs.

Core Characteristics:

- **Hybrid Reasoning**: Often combines deterministic rules with probabilistic or "fuzzy" elements from AI models.
- **Data Flexibility**: Capable of handling both structured and unstructured data, such as text, images, or sensor inputs.
- **Natural Language Processing**: Can process and generate natural language when LLMs or similar models are involved.

Additional Features:

- **Model Integration**: May integrate multiple AI models or services (not limited to LLMs) for various subtasks.
- **Semi-Structured Flows**: Typically designed as modular workflows where different steps (some rule-based, some AI-driven) are chained together.
- **Auditability**: Can include logging and audit trails to track decisions made at each step.
- **Feedback Loops:** May incorporate mechanisms to learn from feedback (although this is often more about retraining the

underlying models off-line rather than “on the fly”).

- **Human-in-the-Loop**: Often designed so that humans can intervene or review decisions when necessary.
- **Exception Handling**: Built with fallback mechanisms to handle cases where AI components may be uncertain or fail.

AI Agents

Definition:
An AI Agent is a system designed to operate autonomously in dynamic environments, making adaptive, non-deterministic decisions. Modern AI Agents often represent an evolution of traditional Large Language Models (LLMs) by extending their capabilities well beyond simple text generation.

Core Characteristics:

- **Adaptive Reasoning:** Uses probabilistic reasoning and machine learning to adjust its actions based on context, goals, and past experiences.
- **Autonomy:** Operates with minimal human intervention, making independent decisions in complex and uncertain environments.

Advanced Capabilities:

- **Enhanced Problem Solving:**
    - **Beyond Text Generation:** While standard LLMs primarily generate text, AI Agents can solve complex, multi-faceted problems that may involve planning and execution in diverse domains.
    - **Strategic Decision-Making:** They can evaluate context and long-term goals to make strategic decisions rather than simply providing reactive responses.
- **Tool and API Integration:**
    - **Interacting with the World:** AI Agents can use external tools and APIs to fetch data, execute commands, and interact with other systems. This capability effectively gives them “hands” for acting in the real world.
- **Learning and Adaptation:**
    - **Continuous Improvement:** They can learn from past interactions and feedback, allowing them to refine their performance over time.
- **Embodied Intelligence:**
    - **“Brain and Hands” Analogy:** Think of an AI Agent as an LLM equipped with a “brain” for reasoning and “hands” for interacting with the external world, enabling both intelligent deliberation and practical action.

Additional Features:

- **Context Awareness:** Considers both immediate and historical context to inform decision-making.

- **Planning and Multi-Step Execution:** Capable of devising and executing multi-step plans to achieve defined objectives.
- **Collaboration:** Can coordinate with other agents or systems to accomplish complex tasks.
- **Handling Uncertainty:** Designed to operate effectively even when facing ambiguous or incomplete information.

21A.25.040 Densities and dimensions – Commercial zones.

A. Commercial Zones.

	ZONES	COMMERCIAL		
		NEIGHBORHOOD BUSINESS	COMMUNITY BUSINESS	OFFICE
STANDARDS		**NB**	**CB**	**O**
Maximum Density DU/Acre		8 du/ac (1)	18 du/ac (1)	18 du/ac (1)
Minimum Lot Area				
Maximum Lot Depth/Width Ratio			10 ft	10 ft
Minimum Street Setback		10 ft (2)	10 ft (2)	10 ft
Minimum Interior Setback (4)		20 ft (5)	20 ft (5)	20 ft (5)
Base Height (7)		35 ft 45 ft	35 ft 60 ft	45 ft 60 ft

Example table

This is an example of a data table.

Disability Category	Participants	Ballots Completed	Ballots Incomplete/ Terminated	Results	
				Accuracy	Time to complete
Blind	5	1	4	34.5%, n=1	1199 sec, n=1
Low Vision	5	2	3	98.3% n=2 (97.7%, n=3)	1716 sec, n=3 (1934 sec, n=2)
Dexterity	5	4	1	98.3%, n=4	1672.1 sec, n=4
Mobility	3	3	0	95.4%, n=3	1416 sec, n=3

Invoice 2022435

Tax invoice

Your Business Name

BILL TO
Your Client
100 Harris St
Sydney NSW NSW 2009
Australia

Issue date: **19/7/2022**
Due date: **3/8/2022**

Reference: **2022435**

Invoice No.	Issue date	Due date	Total due (AUD)
2022435	19/7/2022	3/8/2022	$2,510.00

Description	Quantity	Unit price ($)	Amount ($)
Services & products	1	100.00	100.00
More services & products	1	2,000.00	2,000.00

Subtotal:	$2,100.00
GST 10% *from $100.00*	$10.00
GST 20% *from $2,000.00*	$400.00
Total (AUD):	**$2,510.00**

Issued by, signature:



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Sydney NSW 2000
Australia